

# Supplementary Information

Geographic and Linguistic Bias in LLMs for Air Pollution Policy:  
A 25-Country, 14-Model, 4-Language, 5-Task Benchmark

**Reproducibility.** Every data-driven table below (Tables [S1–S7](#)) is regenerated from the raw confirmatory artifacts by `code/analysis/make_supplement_tables.py`; no value is hand-entered. The primary judge for all reported composites is `gpt-5-mini-2025-08-07`. The analysed confirmatory corpus comprises 9,251 judge-scored responses (7,580 English-prompt; 1,671 native-language) over 25 countries and 14 deployment stacks. All inputs, the scoring code, the ground-truth registry, and the anonymised responses will be released on publication under CC-BY-4.0 (data/prompts) and MIT (code).

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## S1 Pre-registration, design, and country selection

The hypotheses, primary models, persona protocol, exclusion criteria, multiple-comparison corrections, and the country sampling frame were specified in advance of confirmatory data collection. The pre-registered design covers **15 countries**; the sample was subsequently extended, *post registration*, to **25 countries** to gain power for the development gradient (H1) and the corpus-mechanism tests (H4) and to multiply the within-country language pairs (H2). Throughout the

manuscript and this supplement, every effect computed on the 25-country sample is reported alongside its pre-registered 15-country value, and no pre-registered conclusion is revised on the basis of the extension alone (Section S8).

Countries were selected by theoretical sampling triangulated across three independent classifications: the UNCTAD Global North/South classification (operationalising the coloniality-of-knowledge framing), the Joshi et al. (2020) six-class linguistic-resource taxonomy [1], and World Bank income groups. The pre-registered 15 (Set “P” in Table S4) span four world regions, five Joshi classes, and four income groups; the 10 extension countries (Set “E”) comprise seven additional Global North references (United Kingdom, Canada, Australia, South Korea, France, Italy, Portugal) chosen to thicken the high-development end of the gradient, and three additional Global South native-language-pair countries (Colombia, Chile, Angola). The full design rationale is in docs/supplement/s1\_design\_rationale.md.

## S2 Deployment-stack roster and serving configuration

**Unit of analysis.** We evaluate each model *as served* through a fixed access stack — the tuple of model weights, vendor, serving infrastructure, and API/endpoint layer — rather than idealized weights in isolation, because a public manager interacts with the served stack and any measured gap is a property of that stack. Table S1 lists the 14 stacks with their exact API/tag strings and execution venues.

**Table S1.** Deployment-stack roster. Each model is evaluated *as served* through a fixed access stack; the API/tag string and execution venue are part of what each label denotes. Parameter counts are vendor-reported (dense or total for MoE; closed-frontier counts undisclosed). *N* and mean are realized English-prompt confirmatory coverage.

| Model            | Vendor    | Tier | API / tag string                          | Venue            | <i>N</i> | Mean  |
|------------------|-----------|------|-------------------------------------------|------------------|----------|-------|
| DeepSeek-V3      | DeepSeek  | A    | deepseek-chat                             | DeepSeek API     | 500      | 0.625 |
| Command R+       | Cohere    | A    | command-r-plus-08-2024                    | Cohere (trial)   | 500      | 0.509 |
| Llama 3.3 70B    | Meta      | A    | llama-3.3-70b-versatile                   | Groq (free)      | 474      | 0.506 |
| GPT-OSS 120B     | OpenAI    | A    | openai/gpt-oss-120b                       | Groq (free)      | 298      | 0.485 |
| Llama 4 Scout    | Meta      | A    | meta-llama/llama-4-scout-17b-16e-instruct | Groq (free)      | 500      | 0.473 |
| Qwen3 32B        | Alibaba   | B    | qwen/qwen3-32b                            | Groq (free)      | 498      | 0.546 |
| Phi-4 14B        | Microsoft | B    | phi4                                      | Ollama (local)   | 500      | 0.485 |
| Qwen3 14B        | Alibaba   | B    | qwen3:14b                                 | Ollama (local)   | 500      | 0.474 |
| Llama 3.1 8B     | Meta      | C    | llama-3.1-8b-instant                      | Groq (free)      | 777      | 0.424 |
| Cabra-Mistral 7B | botbot-ai | C    | cabra-mistral-7b                          | Ollama (local)   | 496      | 0.320 |
| GPT-5-mini       | OpenAI    | D    | gpt-5-mini                                | OpenAI API       | 777      | 0.626 |
| Gemini 2.5 Flash | Google    | D    | gemini-2.5-flash                          | Google AI (free) | 490      | 0.604 |
| Claude 4.5 Haiku | Anthropic | D    | claude-haiku-4-5                          | Anthropic API    | 490      | 0.597 |

| Model | Vendor | Tier | API / tag string | Venue      | $N$ | Mean  |
|-------|--------|------|------------------|------------|-----|-------|
| GPT-5 | OpenAI | E    | gpt-5            | OpenAI API | 780 | 0.647 |

**Determinism and seed handling.** All stacks were queried at temperature 0.3 with the maximum-determinism configuration each provider documents, not an identical parameter surface, because the surfaces differ. Seed support is heterogeneous: locally served stacks (Ollama) honour a fixed seed; OpenAI exposes an advisory seed; Anthropic’s Messages API exposes no seed parameter; and for the remaining hosted endpoints a seed is either advisory or not exposed. For every call we record the requested seed and a `seed_status` field (`sent`, `logged-only`, or `not-supported`) in the released run metadata, so that the determinism actually achieved is auditable rather than assumed. Two stochastic replications per (prompt, model, persona) cell estimate within-cell variability.

**Acquisition window.** Served-model behaviour can drift as vendors update infrastructure, so the reported accuracies are point-in-time estimates relative to the collection window. The pinned tag, version string, and acquisition timestamp are recorded for every call. Confirmatory collection ran from **2026-04-26 to 2026-06-12** (UTC), spanning the pilot, the pre-registered 15-country collection, and the post-registration extension to 25 countries. Table S2 gives the first and last non-error call timestamp per provider together with the official API documentation consulted; the full per-call timestamps are released with the dataset.

**Table S2.** Per-provider acquisition window and official API documentation. Reported accuracies are point-in-time estimates relative to this window; first/last are the earliest and latest non-error call timestamps (UTC) recorded in the run logs. The window spans the pilot, the pre-registered 15-country collection, and the post-registration extension.

| Provider (venue) | First<br>(UTC) | Last (UTC) | API documentation                                                                                                           |
|------------------|----------------|------------|-----------------------------------------------------------------------------------------------------------------------------|
| Anthropic API    | 2026-04-27     | 2026-06-12 | <a href="https://docs.anthropic.com/en/api/messages">https://docs.anthropic.com/en/api/messages</a>                         |
| Cohere (trial)   | 2026-04-26     | 2026-06-11 | <a href="https://docs.cohere.com/reference/chat">https://docs.cohere.com/reference/chat</a>                                 |
| DeepSeek API     | 2026-04-26     | 2026-06-11 | <a href="https://api-docs.deepseek.com">https://api-docs.deepseek.com</a>                                                   |
| Google AI (free) | 2026-04-27     | 2026-06-11 | <a href="https://ai.google.dev/api">https://ai.google.dev/api</a>                                                           |
| Groq (free)      | 2026-04-26     | 2026-06-11 | <a href="https://console.groq.com/docs/api-reference">https://console.groq.com/docs/api-reference</a>                       |
| Ollama (local)   | 2026-04-27     | 2026-06-12 | <a href="https://github.com/ollama/ollama/blob/main/docs/api.md">https://github.com/ollama/ollama/blob/main/docs/api.md</a> |

| Provider (venue)     | First<br>(UTC) | Last (UTC) | API documentation                                                                                                     |
|----------------------|----------------|------------|-----------------------------------------------------------------------------------------------------------------------|
| OpenAI API           | 2026-04-27     | 2026-06-12 | <a href="https://platform.openai.com/docs/api-reference/chat">https://platform.openai.com/docs/api-reference/chat</a> |
| <b>All providers</b> | 2026-04-26     | 2026-06-12 | —                                                                                                                     |

### S3 Coverage

Per-cell coverage is not uniform across the roster: free and rate-limited endpoints returned fewer admissible responses than the metered APIs, and one candidate (Gemini 2.5 Flash on early native batches) and a sixth judge (Command R+) hit provider quotas. We report realized coverage rather than silently dropping under-covered cells. Table S3 gives the model  $\times$  task English-prompt matrix (column totals equal the by-task counts in the main text); Table S4 gives per-country English, native, and total counts with tier and pre-registration set; Table S5 lists the native-language pairs used for H2. The complete model  $\times$  country  $\times$  task coverage CSV, with the quota or exclusion reason for every shortfall, is included in the release.

**Table S3.** Coverage matrix: realized English-prompt confirmatory responses per model  $\times$  task (T1 standard, T2 local datum, T3 health synthesis, T4 instruments, T5 recommendation). No cell is silently dropped; shortfalls are quota-driven (Methods).

| Model            | T1   | T2   | T3   | T4   | T5   | Total |
|------------------|------|------|------|------|------|-------|
| GPT-5            | 156  | 156  | 156  | 156  | 156  | 780   |
| GPT-5-mini       | 156  | 156  | 156  | 153  | 156  | 777   |
| Llama 3.1 8B     | 156  | 156  | 155  | 156  | 154  | 777   |
| DeepSeek-V3      | 100  | 100  | 100  | 100  | 100  | 500   |
| Command R+       | 100  | 100  | 100  | 100  | 100  | 500   |
| Phi-4 14B        | 100  | 100  | 100  | 100  | 100  | 500   |
| Qwen3 14B        | 100  | 100  | 100  | 100  | 100  | 500   |
| Llama 4 Scout    | 100  | 100  | 100  | 100  | 100  | 500   |
| Qwen3 32B        | 100  | 100  | 100  | 99   | 99   | 498   |
| Cabra-Mistral 7B | 99   | 97   | 100  | 100  | 100  | 496   |
| Gemini 2.5 Flash | 98   | 98   | 98   | 98   | 98   | 490   |
| Claude Haiku 4.5 | 98   | 98   | 98   | 98   | 98   | 490   |
| Llama 3.3 70B    | 98   | 97   | 96   | 92   | 91   | 474   |
| GPT-OSS 120B     | 68   | 72   | 60   | 42   | 56   | 298   |
| <b>Total</b>     | 1529 | 1530 | 1519 | 1494 | 1508 | 7580  |

**Table S4.** Coverage by country: tier (GN/GS), pre-registration status (P = pre-registered 15; E = post-registration extension), and realized confirmatory responses (English, native-language, total).

| Country           | Tier | Set | English | Native | Total |
|-------------------|------|-----|---------|--------|-------|
| Colombia          | GS   | E   | 260     | 251    | 511   |
| Chile             | GS   | E   | 260     | 241    | 501   |
| Portugal          | GN   | E   | 261     | 240    | 501   |
| Angola            | GS   | E   | 260     | 240    | 500   |
| Argentina         | GS   | P   | 355     | 139    | 494   |
| India             | GS   | P   | 352     | 138    | 490   |
| Mexico            | GS   | P   | 329     | 140    | 469   |
| Peru              | GS   | P   | 316     | 141    | 457   |
| Brazil            | GS   | P   | 260     | 141    | 401   |
| Germany           | GN   | P   | 356     | 0      | 356   |
| Indonesia         | GS   | P   | 356     | 0      | 356   |
| Bangladesh        | GS   | P   | 355     | 0      | 355   |
| Egypt             | GS   | P   | 355     | 0      | 355   |
| Kenya             | GS   | P   | 348     | 0      | 348   |
| Japan             | GN   | P   | 347     | 0      | 347   |
| Nigeria           | GS   | P   | 317     | 0      | 317   |
| Philippines       | GS   | P   | 316     | 0      | 316   |
| United States     | GN   | P   | 312     | 0      | 312   |
| South Africa      | GS   | P   | 302     | 0      | 302   |
| United Kingdom    | GN   | E   | 265     | 0      | 265   |
| Canada            | GN   | E   | 260     | 0      | 260   |
| Australia         | GN   | E   | 260     | 0      | 260   |
| France            | GN   | E   | 260     | 0      | 260   |
| Italy             | GN   | E   | 260     | 0      | 260   |
| South Korea       | GN   | E   | 258     | 0      | 258   |
| <b>Total (25)</b> |      |     | 7580    | 1671   | 9251  |

**Table S5.** Native-language confirmatory responses by language and country (H2 within-country English/native contrast).

| Native language (Joshi) | Country  | <i>N</i> (native) |
|-------------------------|----------|-------------------|
| Portuguese (4)          | Portugal | 240               |
|                         | Angola   | 240               |
|                         | Brazil   | 141               |
| Spanish (5)             | Colombia | 251               |
|                         | Chile    | 241               |
|                         | Peru     | 141               |
|                         | Mexico   | 140               |

| Native language (Joshi) | Country   | $N$ (native) |
|-------------------------|-----------|--------------|
|                         | Argentina | 139          |
| Hindi (4)               | India     | 138          |

## S4 Country covariates

Table S6 reports the developmental and corpus-representation covariates used in the mechanism analysis (H4). **Language-corpus** measures (Joshi class; mC4/OSCAR sizes reported in the main text) index how much text exists *in* a country’s language and are the candidate channel for the native-language penalty (H2). **Country-coverage** measures (English-Wikipedia article byte length, Wikidata sitelinks, Wikidata statements) index how broadly the encyclopedic corpus is *about* a country, independent of language, and are the candidate channel for the geographic gap (H1/H4). Because the geographic gap is measured *in English*, a residual North/South gap cannot be a language-corpus effect, which is why H4 isolates country coverage for H1 and language-corpus size for H2. All Wikidata entities are identified by a resolvable QID and were retrieved within the collection window.

**Table S6.** Country covariates. HDI: UNDP HDR 2023–24 (2022 data). Joshi: linguistic-resource class of the dominant official language [1]. Corpus measures from Wikimedia/Wikidata (entity QID resolvable): English-article byte length, Wikidata sitelinks (language editions), and Wikidata statements.

| Country        | Tier | HDI   | Joshi | EN-wiki bytes | Sitelinks | WD stmts |
|----------------|------|-------|-------|---------------|-----------|----------|
| Germany        | GN   | 0.950 | 5     | 207,810       | 403       | 1402     |
| Australia      | GN   | 0.946 | 5     | 233,095       | 395       | 1674     |
| United Kingdom | GN   | 0.940 | 5     | 368,888       | 391       | 1085     |
| Canada         | GN   | 0.935 | 5     | 281,016       | 393       | 1346     |
| South Korea    | GN   | 0.929 | 4     | 275,291       | 350       | 746      |
| United States  | GN   | 0.927 | 5     | 383,914       | 424       | 1710     |
| Japan          | GN   | 0.920 | 5     | 205,844       | 414       | 995      |
| France         | GN   | 0.910 | 5     | 279,498       | 413       | 1196     |
| Italy          | GN   | 0.906 | 4     | 305,234       | 405       | 1039     |
| Portugal       | GN   | 0.874 | 4     | 324,038       | 369       | 751      |
| Chile          | GS   | 0.860 | 5     | 232,108       | 379       | 689      |
| Argentina      | GS   | 0.849 | 5     | 267,662       | 369       | 881      |
| Mexico         | GS   | 0.781 | 5     | 263,757       | 398       | 1286     |
| Peru           | GS   | 0.762 | 5     | 239,544       | 332       | 867      |
| Brazil         | GS   | 0.760 | 4     | 301,844       | 384       | 1151     |
| Colombia       | GS   | 0.758 | 5     | 303,401       | 363       | 774      |
| Egypt          | GS   | 0.728 | 5     | 284,547       | 387       | 757      |
| South Africa   | GS   | 0.717 | 5     | 270,861       | 365       | 750      |
| Indonesia      | GS   | 0.713 | 3     | 226,349       | 362       | 1005     |
| Philippines    | GS   | 0.710 | 5     | 493,981       | 331       | 963      |

| Country    | Tier | HDI   | Joshi | EN-wiki bytes | Sitelinks | WD stmts |
|------------|------|-------|-------|---------------|-----------|----------|
| Bangladesh | GS   | 0.670 | 3     | 275,786       | 343       | 814      |
| India      | GS   | 0.644 | 5     | 310,681       | 399       | 1679     |
| Kenya      | GS   | 0.601 | 5     | 216,189       | 334       | 699      |
| Angola     | GS   | 0.591 | 4     | 180,098       | 322       | 676      |
| Nigeria    | GS   | 0.548 | 5     | 266,041       | 349       | 1194     |

## S5 Ground-truth registry (T1)

The validity of the geographic gradient (H1) requires that ground truth be equally available and equally verifiable across countries. For T1 (the binding annual PM<sub>2.5</sub> standard) we anchored every country to an official primary source by documentary audit: the source text was retrieved, the value was read verbatim, and the entry records the value, status, source-of-record, a resolvable URL, and the retrieval method (Table S7). Three countries have *no* national PM<sub>2.5</sub> standard, confirmed from the official text (Nigeria’s NESREA regulates PM<sub>10</sub> only; Argentina has no binding federal value; Angola has no national air-quality legislation); in those cases a faithful model must decline to state a non-existent threshold rather than fabricate one. This provenance is more reproducible than expert assertion: any reader can re-verify each value from the cited official source. No reference in the registry is unverified, and no value was carried over from secondary sources without confirmation against the primary text.

**Table S7.** Ground-truth registry of official primary sources by document type. VERIFIED\_NO\_STANDARD = verbatim excerpts are in the relea

| Country    | Value                                                                                   |
|------------|-----------------------------------------------------------------------------------------|
| Angola     | NO national air-quality legislation / standard                                          |
| Argentina  | NO single binding federal PM2.5 standard (Ley 20.284/1973 framework; PM2.5 limits set   |
| Australia  | 8 ug/m3 (annual; 7 target 2025)                                                         |
| Bangladesh | 15 ug/m3 (annual) per Gazette S.R.O. 220-Law/2005; a 2022 revision exists but the revis |
| Brazil     | 17 ug/m3 (annual, PI-2)                                                                 |
| Canada     | 8.8 ug/m3 (annual, CAAQS 2020)                                                          |
| Chile      | 20 ug/m3 (annual)                                                                       |

| Country     | Value                                                                                |
|-------------|--------------------------------------------------------------------------------------|
| Colombia    | 25 ug/m3 (annual, in force since 2018; 15 from 2030; 24h=37)                         |
| Egypt       | PM2.5 limit set under Law 4/1994 (amended 9/2009, 105/2015) Executive Regulations (o |
| France      | 25 ug/m3 (annual, in force; 10 from 2030)                                            |
| Germany     | 25 ug/m3 (annual, in force 2026; 10 from 2030)                                       |
| India       | 40 ug/m3 (annual)                                                                    |
| Indonesia   | 15 ug/m3 (annual)                                                                    |
| Italy       | 25 ug/m3 (annual, in force; 10 from 2030)                                            |
| Japan       | 15 ug/m3 (annual)                                                                    |
| Kenya       | 35 ug/m3 (annual; 24h 75)                                                            |
| Mexico      | 10 ug/m3 (annual)                                                                    |
| Nigeria     | NO national PM2.5 standard (PM10 annual=60 ug/m3, 24h=150)                           |
| Peru        | 25 ug/m3 (annual)                                                                    |
| Philippines | 25 ug/m3 (annual, 1-year guideline)                                                  |



| Country        | Value                                     |
|----------------|-------------------------------------------|
| Portugal       | 25 ug/m3 (annual, in force; 10 from 2030) |
| South Africa   | 20 ug/m3 (annual)                         |
| South Korea    | 15 ug/m3 (annual)                         |
| United Kingdom | 10 ug/m3 (annual, target by 2040)         |
| United States  | 9.0 ug/m3 (annual, primary NAAQS, 2024)   |

## S6 Composite outcome and scoring rubric

Every response was scored by the primary judge against the ground-truth registry entry (T1, T2, T4) or the validated rubric (T3, T5), producing five subcomponents each normalised to  $[0, 1]$ :

1. **Factual accuracy** against the registry value (binary for T1; partial credit for T2–T4).
2. **Contextual completeness** scored against a pre-validated rubric (T3, T4).
3. **Citation quality**, verified via DOI/official-source check.
4. **Absence of hallucination**, coded binarily (0 hallucinated, 1 faithful).
5. **Applied validity**, the rubric score for the operational recommendation (T5).

The primary composite uses equal weights. Two pre-specified sensitivity weightings are reported — a PCA-derived set estimated on the Brazil extended pilot, and an author-specified set (factual 0.30, contextual 0.25, citation 0.15, hallucination 0.15, applied 0.15) — and a conclusion is treated as robust only if it holds under all three. The composite is computed per (country, model, persona) cell to support the H6 difference-in-differences test. Each scored record also stores the judge’s free-text rationale; for example, the **gpt-5-mini** judge scored a Llama 4 Scout reply on Brazil’s T1 standard 0.15 with the rationale that it “gives wrong regulation (491/2018 vs correct 506/2024) and wrong current value... included a specific but incorrect citation and fabricated timing,” illustrating how the factual-accuracy and hallucination subcomponents discriminate substantive failure.

## S7 Judge-panel reliability

To address the judge–target overlap (the primary judge, **gpt-5-mini**, is itself among the evaluated tier), a fixed stratified reliability sample of **131 responses** (stratified by country  $\times$  task  $\times$  per-

sona, fixed seed) was independently re-scored by a vendor-diverse panel. The sample was frozen *before* judging — the agreement sampler otherwise re-draws from the live response base, which desynchronised the per-judge samples when the base grew from 15 to 25 countries — and pinned to the judges’ common items by `code/analysis/judge_fixed_keys.py`.

**Table S8.** Inter-judge reliability on the fixed 131-item sample. The operative quantity is the reliability of the panel *mean*,  $\text{ICC}(2, k)$ ; by the Spearman–Brown relation it is high even when single judges agree only moderately. Source: `judge_panel_reliability.json`.

| Quantity                                        | Value                                   |
|-------------------------------------------------|-----------------------------------------|
| Judges ( $k$ )                                  | 4 (OpenAI, Anthropic, Google, DeepSeek) |
| Common items                                    | 131                                     |
| Krippendorff’s $\alpha$ (interval)              | 0.667                                   |
| Single-judge $\text{ICC}(2, 1)$                 | 0.672                                   |
| <b>Panel-mean <math>\text{ICC}(2, 4)</math></b> | <b>0.891</b>                            |
| Pairwise Pearson: GPT-5-mini   Claude Sonnet    | 0.859                                   |
| GPT-5-mini   DeepSeek-V3                        | 0.834                                   |
| Claude Sonnet   DeepSeek-V3                     | 0.821                                   |
| GPT-5-mini   Gemini 2.5 Pro                     | 0.626                                   |
| Claude Sonnet   Gemini 2.5 Pro                  | 0.630                                   |
| DeepSeek-V3   Gemini 2.5 Pro                    | 0.610                                   |

Agreement is strongest among GPT-5-mini, Claude, and DeepSeek (Pearson 0.82–0.86) and lowest for the systematically more lenient Gemini (0.61–0.63); because every reported effect is a between-group contrast (tier gap, language penalty, task floor), a judge’s leniency offset cancels and does not threaten those contrasts. A fifth judge (Llama 3.3 70B, Meta) corroborated on the items it covered (five-judge  $\text{ICC}(2, 5) = 0.861$ ) but, being a noisier 70B judge, *lowered* the panel  $\alpha$  to 0.538 — empirically confirming that four strong judges are preferable to five with a weaker one. A sixth candidate (Command R+, Cohere) was excluded when its trial quota was exhausted mid-run; as an out-of-sample cross-check, Command R+ on the 315 items it did cover agreed with the primary judge at Pearson 0.647 /  $\text{ICC}(2, 1) = 0.602$ . The study uses the single `gpt-5-mini` composite for all effects, validated by this panel, rather than a judge-averaged score; the absence of an independent human-gold layer is stated as a limitation in the main text.

## S8 Statistical methods and full confirmatory test outputs

The three primary confirmatory tests form a single family (F1) corrected together with the Bonferroni–Holm step-down procedure ( $\alpha = 0.05$ ). Secondary confirmatory tests are reported unadjusted within family; declared exploratory tests use Benjamini–Hochberg FDR at  $q = 0.10$ . Rank correlations use Spearman  $\rho$ ; monotonic-trend robustness uses the Mann–Kendall test; the persona effect uses a country-level permutation test (5,000 permutations). All outputs below are regenerated by `code/analysis/formal_tests.py`, `robust_tests.py`, and `h4_corpus_mechanism.py`.

**Table S9.** Primary family (Bonferroni–Holm): pre-registered 15-country values beside the post-registration 25-country extension. H1-by-HDI is the one primary test that rejects its null at  $n = 25$ ; it remains below the pre-registered  $\rho \geq 0.55$  effect-size threshold in both samples.

| Test                                                  | 15 countries (pre-registered)       | 25 countries (extension)                         |
|-------------------------------------------------------|-------------------------------------|--------------------------------------------------|
| H1 Spearman $\rho(\text{acc}, \text{HDI})$            | +0.143 (one-sided $p = 0.306$ ), ns | <b>+0.512</b> ( $p = 0.004$ ), rejects under B–H |
| H1 Mann–Kendall (by HDI)                              | $S = 9$ , $Z = +0.40$ , $p = 0.692$ | $S = 102$ , $Z = +2.36$ , $p = 0.018$            |
| H1 Spearman $\rho(\text{acc}, \text{Joshi})$          | +0.028, ns                          | −0.061, ns (development, not linguistic)         |
| H1 vs threshold $\rho \geq 0.55$                      | not met                             | not met (significant, below effect size)         |
| H4 partial $\rho(\text{acc}, \text{Wiki} \text{HDI})$ | +0.218 ( $p = 0.45$ ), ns           | +0.036 ( $p = 0.87$ ), ns                        |
| H6 DiD persona (GS–GN)                                | +0.97 pp, perm $p = 0.142$ , ns     | +0.39 pp, perm $p = 0.257$ , ns                  |

**Table S10.** H4 corpus mechanism, multi-measure (25 countries). Country-coverage breadth (Wikidata sitelinks) predicts accuracy at the zero-order level and survives multiplicity correction across the three coverage proxies, but attenuates to non-significance after adjusting for development (HDI); we therefore report it as suggestive/exploratory, not as an established mechanism. The pre-registered language-edition proxy is null.

| Measure                   | Family   | $\rho(\text{acc}, \text{log measure})$ | partial   HDI         |
|---------------------------|----------|----------------------------------------|-----------------------|
| English-Wikipedia bytes   | country  | +0.198 ( $p = 0.34$ )                  | +0.181                |
| <b>Wikidata sitelinks</b> | country  | <b>+0.539</b> ( $p = 0.005$ )          | +0.317 ( $p = 0.13$ ) |
| Wikidata statements       | country  | +0.313 ( $p = 0.13$ )                  | +0.144                |
| Wikipedia-edition count   | language | +0.082, ns                             | +0.036, ns            |

## S9 Prompt construction, persona template, and verbatim examples

Prompts were authored in English for the air pollution policy domain with domain supervision (PPGSAU/UTFPR) and validated against the ground-truth registry before entering the confirmatory stimulus set; a pilot subset met an inter-rater threshold of Krippendorff’s  $\alpha \geq 0.70$  before scaling. Native-language renderings (Portuguese, Spanish, Hindi) were translated and back-translated by native speakers; any prompt whose back-translation produced a semantic shift was rewritten or excluded. Every factual claim in a ground-truth entry was required to resolve to an official-source URL or a peer-reviewed DOI before the corresponding prompt was admitted (the project Citation Verification Protocol applied at the ground-truth generation stage, not only at the manuscript stage).

**Persona manipulation.** Persona is a within-prompt factor with two levels that hold the substantive question identical. The **neutral** level presents a plain information request; the **public\_manager\_env** level prepends a fixed role frame. The contrast is shown verbatim for Brazil, T1:

**Neutral.** “What is the current national annual mean ambient air quality standard for PM2.5 in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences.”

**Public environmental manager.** “*I am the municipal environmental management official of São Paulo, Brazil, and I need to brief decision-makers.* What is the current national annual mean ambient air quality standard for PM2.5 in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences.”

**Table S11.** Robust secondary findings (25 countries), with leave-one-country-out (LOCO) and range-restriction checks.

| Finding                                          | Statistic                                                               |
|--------------------------------------------------|-------------------------------------------------------------------------|
| Tier gap (GN–GS)                                 | +6.2 pp, 95% CI [+3.7, +8.6] (15-country: +6.7 [+2.8, +10.2])           |
| Tier gap, LOCO range                             | [+5.8, +7.1] pp, stable                                                 |
| H1 HDI gradient, LOCO range                      | $\rho \in [+0.48$ (drop NGA), +0.66 (drop IND)]                         |
| H1 range-restricted (within 15-country HDI span) | $\rho = 0.512$ ( $p = 0.009$ ): density, not range                      |
| Task floor (T1+T2 vs T3–T5)                      | 0.367 vs 0.638, Mann–Whitney, Cliff’s $\delta = -0.64$                  |
| H3 Cabra–Mistral vs rest                         | 0.320 vs 0.543, Cliff’s $\delta = -0.51$                                |
| H2 native vs English (all)                       | $\Delta = -2.1$ pp, Wilcoxon $p = 1.96 \times 10^{-3}$ ( $n = 762$ )    |
| Spanish / Portuguese / Hindi                     | $-1.0$ ( $p = 0.15$ ) / $-2.3$ ( $p = 0.033$ ) / $-7.8$ ( $p = 0.013$ ) |
| H2 country-level (independent unit, $n = 9$ )    | 9/9 negative, mean $-2.4$ pp, Wilcoxon $p = 0.008$                      |
| H5 open vs closed-accessible                     | +14.1 pp closed advantage (descriptive)                                 |

The corresponding registry ground truth is: “Brazil’s national PM<sub>2.5</sub> standards are set by CONAMA Resolution 506/2024 (which superseded the relevant articles of Resolution 491/2018). The stage in force in 2026 is Intermediate Target PI-2; the Final Target adopts the WHO 2021 annual guideline of 5  $\mu\text{g}/\text{m}^3$ ” (source: CONAMA Resolution 506/2024). Native-language prompts apply the same template, translated; for example the Hindi T1 prompt for India is the faithful Devanagari rendering of the neutral English item, reproduced verbatim in the released prompt set. The complete prompt set (370 confirmatory items: 25 countries  $\times$  5 tasks  $\times$  persona, plus native-language pairs) is released as `prompts_confirmatory.jsonl` with the per-item ground truth, source, and rubric.

## References

- [1] Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.560.